

Refresher Course in Calculus, Probability, and Statistics

Day 2: Optimization

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Introduction

- ▶ **Optimization:** the process of finding the relative maximum or minimum of a function.
 - ▶ Optimization problems are among the most frequent in economics and finance:
 - maximize profit or minimize cost (producer)
 - maximize utility (consumer)
 - maximize return or minimize variance (investor)
 - minimize the sum of squared errors when estimating a regression
 - ...
 - ▶ In economics and finance, optimization problems often imply constraints
- ⇒ **Constrained optimization.**

Optimization

- Requires primarily a (relevant) **objective function**
 - functional relationship between the object of optimization and its determinants

$$\underbrace{y}_{\text{object of optimization}} = \underbrace{f}_{\text{objective function}}(\underbrace{x}_{\substack{\text{choice variable(s)} \\ \downarrow}})$$

- usually assumed to be (twice) continuously differentiable
- Example: maximize profit π
 - profit is the difference between total revenue R and total cost C
 - both revenue and cost are functions of the quantity produced Q :
 $R = R(Q), \quad C = C(Q)$
 - objective function of the profit-maximization problem:

$$\pi(Q) = R(Q) - C(Q)$$

- the only choice variable is the quantity Q

First-order condition (FOC)

To find local/global **extrema** of a function, take the first derivative and set it equal to zero:

$$f'(x) = 0$$

- ▶ Necessary condition, known as the **first-order condition (FOC)**.
Necessary but *not* sufficient.
- ▶ Identifies all points at which the function is neither increasing nor decreasing, but at a plateau.
- ▶ All points identified this way are **candidates** (or critical points) for a possible maximum or minimum.

Unconstrained Optimization (continued)

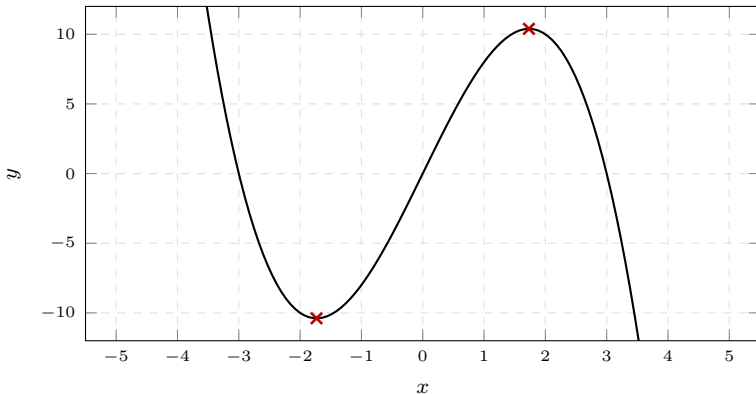
Second-order condition

- ▶ Once critical points are identified by the FOC (i.e. $f'(x) = 0$ for some $x = a$):
 - take the second derivative of $f(x)$
 - evaluate it at each critical point
- ▶ Assuming the necessary FOC is met, the **second-order condition (SOC)** is a sufficient condition to qualify an extremum:

If...	then the function is...	and a is a...
$f''(a) < 0$	concave at a	relative maximum
$f''(a) > 0$	convex at a	relative minimum
$f''(a) = 0$	test is inconclusive	

Unconstrained Optimization (continued)

Example: find the extreme points, maximum(s) and minimum(s) of $f(x) = -x^3 + 9x$

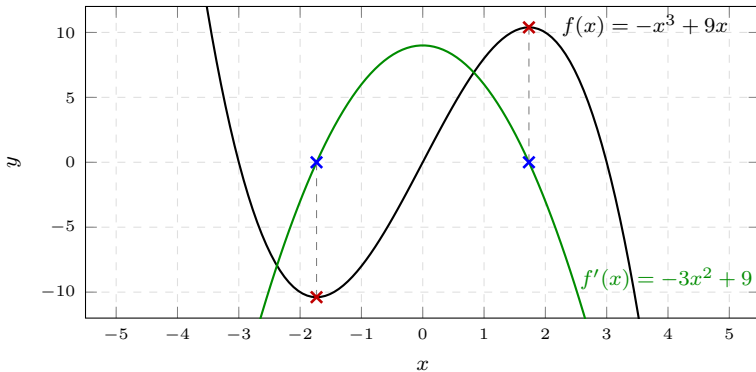


Unconstrained Optimization (continued)

► FOC: $f'(x) = 0 = -3x^2 + 9$

◦ roots: $x_{1,2} = \pm\sqrt{3}$

◦ extreme points: $(x_1^*, y_1^*) = (\sqrt{3}, 6\sqrt{3})$ and $(x_2^*, y_2^*) = (-\sqrt{3}, -6\sqrt{3})$



NB: $f(\sqrt{3}) = -3\sqrt{3} + 9\sqrt{3} = 6\sqrt{3}$; $f(-\sqrt{3}) = 3\sqrt{3} - 9\sqrt{3} = -6\sqrt{3}$.

Unconstrained Optimization (continued)

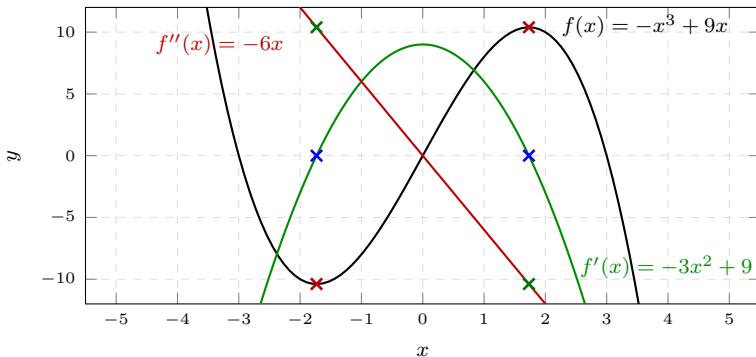
► SOC: $f''(x) = -6x$

◦ at $x = x_1^* = \sqrt{3}$: $f''(\sqrt{3}) = -6\sqrt{3} < 0$

⇒ $(x_1^*, y_1^*) = (\sqrt{3}, 6\sqrt{3})$ is a **maximum**.

◦ at $x = x_2^* = -\sqrt{3}$: $f''(-\sqrt{3}) = 6\sqrt{3} > 0$

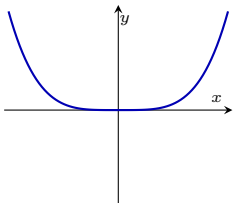
⇒ $(x_2^*, y_2^*) = (-\sqrt{3}, -6\sqrt{3})$ is a **minimum**.



Unconstrained Optimization (continued)

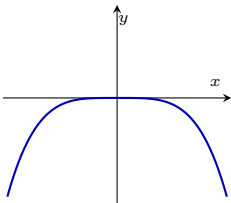
Possible cases where $f'(a) = f''(a) = 0$ (here with $a = 0$)

$$y = x^4$$



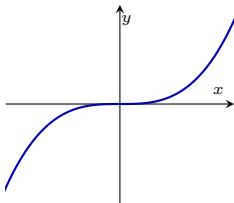
Minimum

$$y = -x^4$$



Maximum

$$y = x^3$$



Inflection point

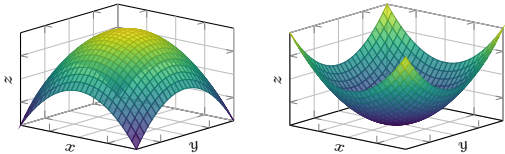
Unconstrained Optimization (continued)

Successive-derivative test

- ▶ If $f''(a) = 0$, the SOC is inconclusive.
- ▶ The successive-derivative test is helpful: evaluate higher-order derivatives at the critical point.
- ▶ If:
 - the 1st non-zero higher-order derivative is an **odd**-numbered derivative (3rd, 5th, etc.):
 - **inflection point**, i.e. a point at which the function changes from being convex ($f''(a) > 0$) to concave ($f''(a) < 0$), or vice versa
 - the 1st non-zero higher-order derivative is an **even**-numbered derivative (4th, 6th, etc.) and the value of this derivative is:
 - negative $\Rightarrow$ **maximum**
 - positive $\Rightarrow$ **minimum**

Multivariable Functions (continued)

- Searching for extrema of a multivariable function $z = f(x, y)$:

Condition(s)	Maximum	Minimum
First-order necessary	$\frac{\partial f(x,y)}{\partial x} = f_x = 0$ and $\frac{\partial f(x,y)}{\partial y} = f_y = 0$	
Second-order necessary	$\frac{\partial^2 f}{\partial x^2} = f_{xx} < 0$ and $\frac{\partial^2 f}{\partial y^2} = f_{yy} < 0$	$\frac{\partial^2 f}{\partial x^2} = f_{xx} > 0$ and $\frac{\partial^2 f}{\partial y^2} = f_{yy} > 0$
Second-order sufficient	$f_{xx} \times f_{yy} > (f_{xy})^2$	
Shape		

NB: the sufficient SOC follows from: $\det \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix} = f_{xx}f_{yy} - (f_{xy})^2 > 0$.

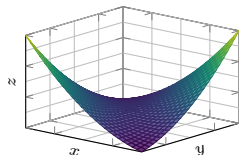
Multivariable Functions (continued)

► Searching for extrema of a multivariable function $z = f(x, y)$. If:

- $f_{xx} \times f_{yy} < (f_{xy})^2$ and f_{xx} & f_{yy} same signs

→ **inflection point**

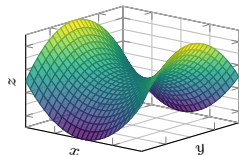
e.g. $z = x^2 + y^2 + 3xy$



- $f_{xx} \times f_{yy} < (f_{xy})^2$ and f_{xx} & f_{yy} opposite signs

→ **saddle point**

e.g. $z = x^2 - y^2$



- $f_{xx} \times f_{yy} = (f_{xy})^2$: test is inconclusive

e.g. $z = x^3 + y^3$

NB: 3D plots can be generated at <https://www.geogebra.org/3d?lang=fr>

Multivariable Functions (continued)

Example 1: Consider: $f(x, y) = 3x^2 - xy + 2y^2 - 4x - 7y + 12$

► FOCs:

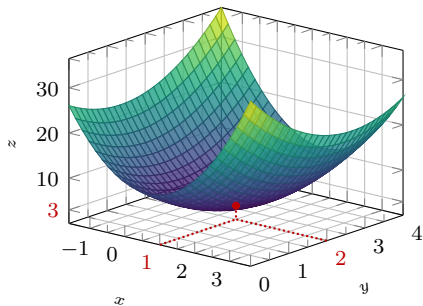
- $f_x = 6x - y - 4 = 0$
- $f_y = -x + 4y - 7 = 0$
- one critical point at $(x^*, y^*) = (1, 2)$

► SOC:

- $f_{xx} = 6$
- $f_{yy} = 4$
- $f_{xy} = f_{yx} = -1$

$$\rightarrow f_{xx} \times f_{yy} = 24 > (f_{xy})^2 = (-1)^2 = 1$$

$\Rightarrow f(x, y)$ has an extremum at $(x^*, y^*) = (1, 2)$, which is a **minimum**.



Multivariable Functions (continued)

Example 2: $f(x, y) = 3x^2y + y^3 - 3x^2 - 3y^2 + 2$

Strategy: solve the system $f_x = 0$, $f_y = 0$ by factoring the equation:
a product is zero exactly when one of its factors is, so the factored equation splits every solution into exhaustive cases.

► **FOCs:** $f_x = 6xy - 6x = 6x(y - 1)$, $f_y = 3x^2 + 3y^2 - 6y$.

- **Step 1 – factor and split:** $6x(y - 1) = 0$ holds only for $x = 0$ or $y = 1$.
- **Step 2 – substitute each case into $f_y = 0$,** i.e. $x^2 + y^2 - 2y = 0$:

$$x = 0 : y^2 - 2y = y(y - 2) = 0 \Rightarrow y = 0 \text{ or } y = 2$$

$$y = 1 : x^2 - 1 = 0 \Rightarrow x = \pm 1$$

► **Critical points** (union of both cases): $(0, 0)$, $(0, 2)$, $(1, 1)$, $(-1, 1)$.

Multivariable Functions (continued)

Example 2 continued:

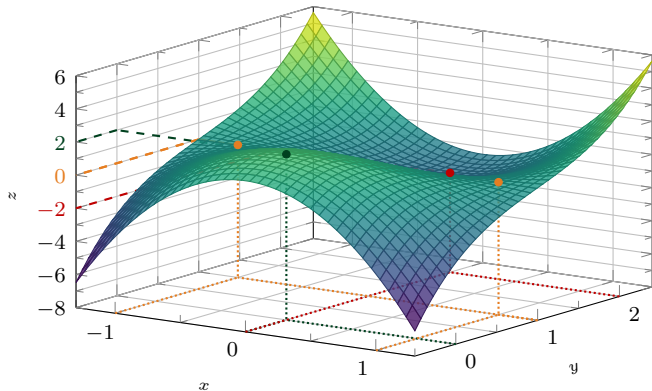
- SOCs at each of the critical points $(0,0)$, $(0,2)$, $(1,1)$, $(-1,1)$:

$$f_{xx} = 6y - 6, \quad f_{yy} = 6y - 6, \quad f_{xy} = f_{yx} = 6x$$

- at $(0,0)$: $f_{xx} = f_{yy} = -6$, $f_{xy} = 0$
→ $f_{xx}f_{yy} = 36 > 0$: **maximum**
- at $(0,2)$: $f_{xx} = f_{yy} = 6$, $f_{xy} = 0$
→ $f_{xx}f_{yy} = 36 > 0$: **minimum**
- at $(1,1)$: $f_{xx} = f_{yy} = 0$, $f_{xy} = 6$
→ $f_{xx}f_{yy} = 0 < 36$: **saddle point**
- at $(-1,1)$: $f_{xx} = f_{yy} = 0$, $f_{xy} = -6$
→ $f_{xx}f_{yy} = 0 < 36$: **saddle point**

Multivariable Functions (continued)

Example 2: $f(x, y) = 3x^2y + y^3 - 3x^2 - 3y^2 + 2$



Constrained Optimization

Simple problem: utility maximization

$$\begin{array}{lll} \text{Max/Min} & \underbrace{U = xy + 2x}_{\substack{\text{objective function} \\ \text{e.g., utility function}}} & \text{subject to (s.t.)} \quad \underbrace{4x + 2y = 60}_{\text{budget constraint}} \end{array}$$

Note:

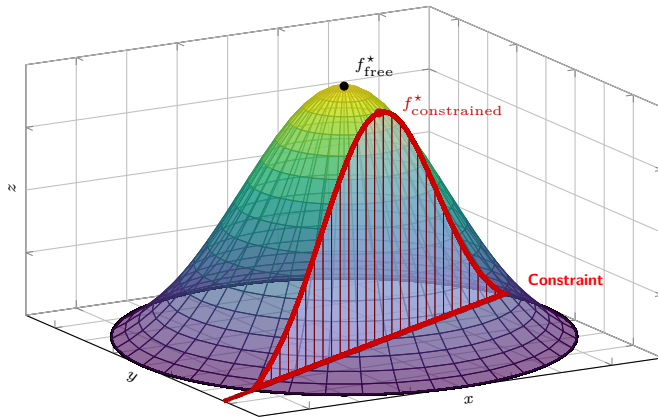
- $\frac{\partial U}{\partial x} > 0$ and $\frac{\partial U}{\partial y} > 0$ (for $x, y \in \mathbb{R}^+$) $\rightarrow$ to maximize U without constraints, the consumer needs to buy an *infinite* amount of both goods
- the budget constraint restricts the problem, taking into account the wealth of the consumer.

Simple solution:

- ▶ use the restriction to find $y = \frac{60 - 4x}{2} = 30 - 2x$
- ▶ substitute in $U = x(30 - 2x) + 2x = -2x^2 + 32x$
- ▶ solve the single-variable problem:
 - FOC: $\frac{\partial U}{\partial x} = -4x + 32 = 0 \rightarrow x = 8$ and $y = 14$
 - SOC: $\frac{\partial^2 U}{\partial x^2} = -4 < 0$: objective **maximized** at $(x^*, y^*) = (8, 14)$

Constrained Optimization

Illustration of a function with two variables $z = f(x, y)$



Schematic after Chiang (1984: 371)

Constrained Optimization

The Lagrange multiplier method

- ▶ A solution to a problem

$$\text{Max/Min } f(x_1, x_2) \quad \text{s.t.} \quad g(x_1, x_2) = c$$

- ▶ can be obtained by:

1. Write down the Lagrangian function:

$$\mathcal{L}(x_1, x_2, \lambda) = f(x_1, x_2) + \lambda(c - g(x_1, x_2))$$

where λ is the **Lagrange multiplier**.

2. Differentiate $\mathcal{L}$ wrt x_1 , x_2 , and λ , and set all to 0 (FOCs):

$$\frac{\partial \mathcal{L}}{\partial x_1} = 0, \quad \frac{\partial \mathcal{L}}{\partial x_2} = 0, \quad \frac{\partial \mathcal{L}}{\partial \lambda} = c - g(x_1, x_2) = 0$$

3. Solve for the three unknowns x_1, x_2, λ to find the critical values.

Constrained Optimization

Lagrange multiplier method — Example 1:

$$\max U = xy + 2x \quad \text{s.t.} \quad 4x + 2y = 60$$

1. Write the Lagrangian: $\mathcal{L}(x, y, \lambda) = xy + 2x + \lambda(60 - 4x - 2y)$
2. Differentiate wrt x, y, λ and set to 0:

$$\frac{\partial \mathcal{L}}{\partial x} = y + 2 - 4\lambda = 0 \quad (1), \quad \frac{\partial \mathcal{L}}{\partial y} = x - 2\lambda = 0 \quad (2), \quad \frac{\partial \mathcal{L}}{\partial \lambda} = 60 - 4x - 2y = 0 \quad (3)$$

3. Solve for x, y, λ :

- from 1st eq.: $\lambda = \frac{y}{4} + \frac{1}{2}$; from 2nd eq.: $\lambda = \frac{x}{2}$

→ thus $\frac{y}{4} + \frac{1}{2} = \frac{x}{2} \Leftrightarrow x = \frac{y}{2} + 1.$

- in 3rd eq.: $4(\frac{y}{2} + 1) + 2y = 60 \Rightarrow y = 14$

- then $x = 8$, and from 1st/2nd eq.: $\lambda = 4.$

⇒ **Critical values:** $(x^*, y^*, \lambda^*) = (8, 14, 4)$

Constrained Optimization (continued)

Significance of the Lagrange multiplier

- ▶ The Lagrange multiplier λ approximates the **marginal impact on the objective function** caused by a small change in the constraint:
 - a 1-unit increase (decrease) in c would cause the optimal value to increase (decrease) by approximately λ units.
 - e.g., $U_{\max}$ should increase by 4 units if the constraint changes from $4x + 2y = 60$ to $4x + 2y = 61$
 - Economic interpretation: when the budget constraint changes by 1 CHF, utility changes by 4 units = the marginal utility of 1 CHF
- ▶ λ is referred to as the **shadow price** of the resource
- ▶ NB: $\mathcal{L}(x_1, x_2, \lambda) = f(x_1, x_2) + \lambda(c - g(x_1, x_2))$ is equivalent to
$$\mathcal{L}(x_1, x_2, \lambda) = f(x_1, x_2) - \lambda(g(x_1, x_2) - c)$$
 - critical values are identical
 - only the sign of λ differs

Constrained Optimization (continued)

Lagrange multiplier method — Example 2:

Optimize $f(x, y) = 4x^2 + 3xy + 6y^2$ s.t. $g(x, y) = x + y = 56$

Set up the Lagrangian:

$$\mathcal{L}(x, y, \lambda) = 4x^2 + 3xy + 6y^2 + \lambda(56 - x - y)$$

Differentiate wrt x, y, λ and set to zero:

$$\frac{\partial \mathcal{L}}{\partial x} = 8x + 3y - \lambda = 0 \quad (1), \quad \frac{\partial \mathcal{L}}{\partial y} = 3x + 12y - \lambda = 0 \quad (2), \quad \frac{\partial \mathcal{L}}{\partial \lambda} = 56 - x - y = 0 \quad (3)$$

Solve for x, y, λ :

- subtract (2) from (1): $5x - 9y = 0 \Rightarrow x = \frac{9}{5}y$
- substitute into (3): $56 - \frac{9}{5}y - y = 0 \Rightarrow y = 20 \Rightarrow x = 36 \Rightarrow \lambda = 348$

Constrained Optimization (continued)

Lagrange multiplier method — Example 2:

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- substitute into (3): $56 - \frac{9}{5}y - y = 0 \Rightarrow y = 20 \Rightarrow x = 36 \Rightarrow \lambda = 348$

Critical values $(x^*, y^*, \lambda^*) = (36, 20, 348) \quad \Rightarrow \quad \text{Min or Max?}$

Constrained Optimization — Key building blocks

1. The Hessian of a function is the (symmetric) matrix of its second-order partial derivatives. For $f(x, y)$:

$$\mathbf{H}(f) = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} \quad (f_{xy} = f_{yx} \text{ when the partials are continuous, by Young})$$

2. The bordered Hessian is simply the *full* Hessian of the Lagrangian $\mathcal{L}(x, y, \lambda)$ including second derivatives w.r.t. *all* variables, with λ listed first:

$$\overline{\mathbf{H}} = \begin{bmatrix} \mathcal{L}_{\lambda\lambda} & \mathcal{L}_{\lambda x} & \mathcal{L}_{\lambda y} \\ \mathcal{L}_{x\lambda} & \mathcal{L}_{xx} & \mathcal{L}_{xy} \\ \mathcal{L}_{y\lambda} & \mathcal{L}_{yx} & \mathcal{L}_{yy} \end{bmatrix} = \begin{bmatrix} 0 & -g_x & -g_y \\ -g_x & \mathcal{L}_{xx} & \mathcal{L}_{xy} \\ -g_y & \mathcal{L}_{yx} & \mathcal{L}_{yy} \end{bmatrix}$$

since $\mathcal{L}_{\lambda\lambda} = 0$ and $\mathcal{L}_{\lambda x} = -g_x$, $\mathcal{L}_{\lambda y} = -g_y$. The “border” is just the constraint g ’s gradient.

3. The determinant of a 3×3 matrix (cofactor expansion along row 1, starting with $a = 0$):

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a(ei - fh) - b(di - fg) + c(dh - eg)$$

Constrained Optimization (continued)

Lagrange multiplier method

Is $(36, 20, 348)$ a minimum or a maximum?

Bordered Hessian test

$$|\overline{\mathbf{H}}(x^*, y^*, \lambda^*)| > 0 \Rightarrow \text{local maximum} \quad |\overline{\mathbf{H}}(x^*, y^*, \lambda^*)| < 0 \Rightarrow \text{local minimum}$$

$$|\overline{\mathbf{H}}| = \begin{vmatrix} 0 & -g_x & -g_y \\ -g_x & \mathcal{L}_{xx} & \mathcal{L}_{xy} \\ -g_y & \mathcal{L}_{yx} & \mathcal{L}_{yy} \end{vmatrix} = -(-g_x)[-g_x\mathcal{L}_{yy} - \mathcal{L}_{xy}(-g_y)] + (-g_y)[-g_x\mathcal{L}_{yx} - \mathcal{L}_{xx}(-g_y)]$$

NB: $\overline{\mathbf{H}}$ is the bordered Hessian *matrix*; $|\overline{\mathbf{H}}|$ is its determinant.

Constrained Optimization (continued)

Lagrange multiplier method

Second-order partials: direct $\mathcal{L}_{xx} = 8$, $\mathcal{L}_{yy} = 12$; cross $\mathcal{L}_{xy} = 3 = \mathcal{L}_{yx}$

$$g(x, y) = x + y = 56 \Rightarrow g_x = 1, g_y = 1$$

Calculate the determinant of the bordered Hessian:

$$\begin{aligned} |\overline{\mathbf{H}}(x^*, y^*, \lambda^*)| &= \begin{vmatrix} 0 & -g_x & -g_y \\ -g_x & \mathcal{L}_{xx} & \mathcal{L}_{xy} \\ -g_y & \mathcal{L}_{yx} & \mathcal{L}_{yy} \end{vmatrix} = \begin{vmatrix} 0 & -1 & -1 \\ -1 & 8 & 3 \\ -1 & 3 & 12 \end{vmatrix} \\ &= -(-1) \left((-1) \times 12 - (-1) \times 3 \right) + (-1) \left((-1) \times 3 - (-1) \times 8 \right) \\ &= -9 + (-5) = -14 < 0. \end{aligned}$$

Result

$(x^*, y^*, \lambda^*) = (36, 20, 348)$ is a **local minimum**.

Constrained Optimization (continued)

Lagrange multiplier method — back to the first example

- Compute the local max/min of $f(x, y) = xy + 2x$ s.t. $g(x, y) = 4x + 2y = 60$

$$\mathcal{L}(x, y, \lambda) = xy + 2x + \lambda(60 - 4x - 2y)$$

→ Critical value: $(x^*, y^*, \lambda^*) = (8, 14, 4)$

- First-order partials:

$$\mathcal{L}_x = y + 2 - 4\lambda \quad \mathcal{L}_y = x - 2\lambda$$

- Second-order partials: direct $\mathcal{L}_{xx} = 0$, $\mathcal{L}_{yy} = 0$; cross $\mathcal{L}_{xy} = 1 = \mathcal{L}_{yx}$

$$g(x, y) = 4x + 2y = 60 \Rightarrow g_x = 4, g_y = 2$$

Constrained Optimization (continued)

Lagrange multiplier method

Calculate the determinant of the bordered Hessian:

$$\overline{\mathbf{H}}(x^*, y^*, \lambda^*) = \begin{vmatrix} 0 & -g_x & -g_y \\ -g_x & \mathcal{L}_{xx} & \mathcal{L}_{xy} \\ -g_y & \mathcal{L}_{yx} & \mathcal{L}_{yy} \end{vmatrix} = \begin{vmatrix} 0 & -4 & -2 \\ -4 & 0 & 1 \\ -2 & 1 & 0 \end{vmatrix} = 16 > 0$$

Result

$(x^*, y^*, \lambda^*) = (8, 14, 4)$ is a **local maximum**.

Constrained Optimization (continued)

Multi-constraint case

- Consider a problem with n variables and m constraints:

$$\underset{x_1, \dots, x_n}{\text{Max/Min}} f(x_1, \dots, x_n) \quad \text{s.t.} \quad \begin{cases} g_1(x_1, \dots, x_n) = c_1 \\ \vdots \\ g_m(x_1, \dots, x_n) = c_m \end{cases}$$

- The Lagrangian is:

$$\mathcal{L}(x_1, \dots, x_n, \lambda_1, \dots, \lambda_m) = f(x_1, \dots, x_n) + \sum_{j=1}^m \lambda_j (c_j - g_j(x_1, \dots, x_n))$$

- The FOCs write:

$$\frac{\partial \mathcal{L}}{\partial x_i} = \frac{\partial f}{\partial x_i} - \sum_{j=1}^m \lambda_j \frac{\partial g_j}{\partial x_i} = 0, \quad \frac{\partial \mathcal{L}}{\partial \lambda_j} = c_j - g_j(x_1, \dots, x_n) = 0$$

Constrained Optimization (continued)

Example: multi-constraint case

Optimize $f(x, y, z) = x^2 + y^2 + z^2$ s.t.

$$\begin{cases} g_1(x, y, z) = x + y + z = 1 \\ g_2(x, y, z) = x + 2y + 3z = 6 \end{cases}$$

► **Step 1:** set up the Lagrangian

$$\mathcal{L}(x, y, z, \lambda_1, \lambda_2) = x^2 + y^2 + z^2 + \lambda_1(1 - x - y - z) + \lambda_2(6 - x - 2y - 3z)$$

► **Step 2:** take the first partial derivatives wrt $x, y, z, \lambda_1, \lambda_2$:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x} = 2x - \lambda_1 - \lambda_2 = 0 & \quad \frac{\partial \mathcal{L}}{\partial y} = 2y - \lambda_1 - 2\lambda_2 = 0 & \quad \frac{\partial \mathcal{L}}{\partial z} = 2z - \lambda_1 - 3\lambda_2 = 0 \\ \frac{\partial \mathcal{L}}{\partial \lambda_1} = 1 - x - y - z = 0 & \quad \frac{\partial \mathcal{L}}{\partial \lambda_2} = 6 - x - 2y - 3z = 0 \end{aligned}$$

Constrained Optimization (continued)

Example: multi-constraint case (continued)

Solving the five-equation system yields the critical values:

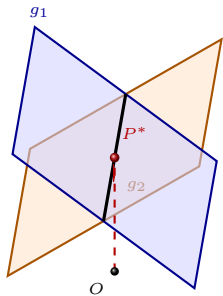
Critical point

$$(x^*, y^*, z^*) = \left(-\frac{5}{3}, \frac{1}{3}, \frac{7}{3}\right), \quad (\lambda_1^*, \lambda_2^*) = \left(-\frac{22}{3}, 4\right)$$

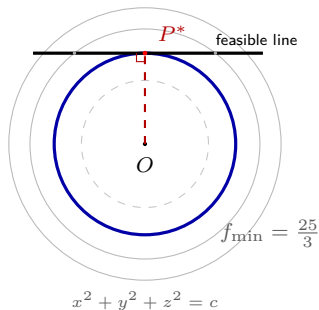
- The objective $f(x, y, z) = x^2 + y^2 + z^2$ is the squared distance from the origin, so this point is the **minimum**-distance point on the intersection of the two constraint planes.

Constrained Optimization — geometric interpretation

Isosurfaces of $f(x, y, z) = x^2 + y^2 + z^2 = c$ are *spheres* in the domain $\mathbb{R}^3$. The minimum is the smallest sphere that still touches the feasible line.



Feasible set = $g_1 \cap g_2$ (a line in $\mathbb{R}^3$)



Isosurfaces of f (spheres), sliced through O ; smallest is tangent at P^*